

$$\begin{array}{ccc}
 A_c(\infty) & \xrightarrow{P_c} & A_c(\infty) \\
 \downarrow \varphi_c & & \downarrow \varphi_c \\
 \mathbb{C} \setminus \overline{\mathbb{D}} & \xrightarrow{z^2} & \mathbb{C} \setminus \overline{\mathbb{D}}
 \end{array}$$